

Flott, höldum áfram í næstu útgáfu "Technical Specification v0.3 – PostgreSQL Schema & API".
Frábært. Þá förum við úr gagnalíkaninu yfir í **raunverulega gagnagrunns- og API-forskrift**.

Technical Specification v0.3

PostgreSQL Schema & REST API

Project: Icelandic Legal Case Management System — ILCMS

Version: 0.3

Status: Technical Design Draft

Database: PostgreSQL 16+

API: REST / JSON

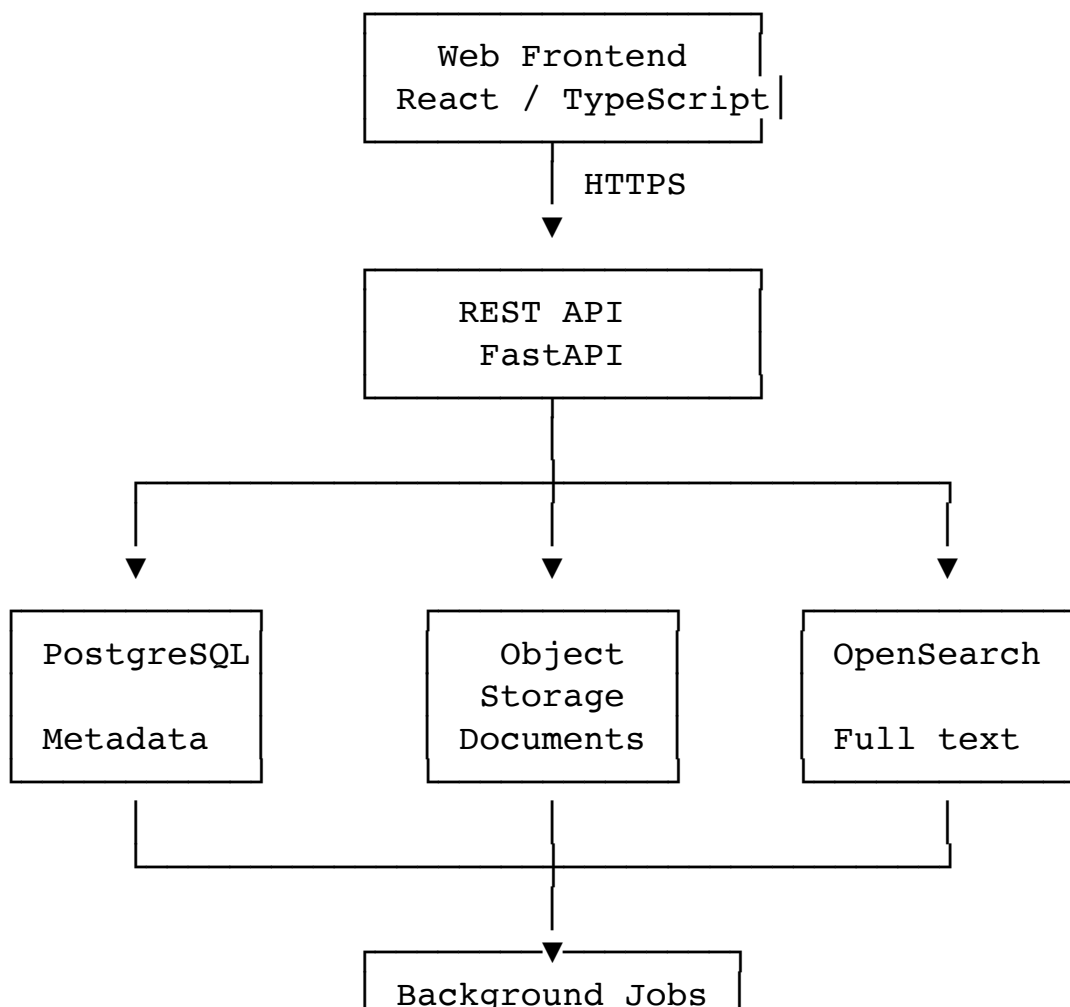
Authentication: OAuth 2.0 / OpenID Connect

Encoding: UTF-8

Time storage: UTC / TIMESTAMPTZ

1. Architecture

Miðað við v0.2 legg ég til þessa uppbyggingu:



Mikilvægt: PostgreSQL er authoritative source fyrir metadata og relational state. Object storage er authoritative source fyrir sjálf skjölin.

2. PostgreSQL grunnstilling

Ég myndi nota:

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;  
CREATE EXTENSION IF NOT EXISTS citext;  
pgcrypto gefur okkur meðal annars gen_random_uuid().
```

`citext` er gagnlegt fyrir email og aðra textareiti þar sem uppercase/lowercase á ekki að skipta máli.

3. Common audit columns

Flestar helstu töflur fá:

```
created_at  
created_by  
updated_at  
updated_by  
Dæmi:
```

```
created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
created_by UUID NOT NULL,  
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
updated_by UUID
```

Ég myndi **ekki** setja `updated_at` á append-only töflur eins og `audit_log` og `evidence_custody_events`.

4. Core tables

4.1 users

```
CREATE TABLE users (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    external_subject VARCHAR(500) NOT NULL UNIQUE,
```

```

username VARCHAR(200),
display_name VARCHAR(500) NOT NULL,
email CTEXT NOT NULL,

status VARCHAR(50) NOT NULL DEFAULT 'ACTIVE',

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
last_login_at TIMESTAMPTZ
);

```

Allowed status:

```

ACTIVE
DISABLED
SUSPENDED

```

5. cases

```

CREATE TABLE cases (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_number VARCHAR(50) NOT NULL UNIQUE,
    title VARCHAR(500) NOT NULL,
    description TEXT,

    case_type_id UUID,

    status VARCHAR(50) NOT NULL DEFAULT 'NEW',
    priority VARCHAR(20) DEFAULT 'NORMAL',

    opened_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    closed_at TIMESTAMPTZ,

    responsible_user_id UUID,

    confidentiality_level_id UUID,
    retention_policy_id UUID,

    created_by UUID NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_by UUID,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    CONSTRAINT cases_status_chk
        CHECK (status IN (

```

```

        'NEW',
        'OPEN',
        'ON_HOLD',
        'PRE_LITIGATION',
        'LITIGATION',
        'APPEAL',
        'JUDGMENT',
        'CLOSED',
        'ARCHIVED'
    ))
);
Indexes:

```

```

CREATE INDEX idx_cases_status
    ON cases(status);

```

```

CREATE INDEX idx_cases_responsible
    ON cases(responsible_user_id);

```

```

CREATE INDEX idx_cases_opened_at
    ON cases(opened_at);

```

6. Persons

```

CREATE TABLE persons (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    first_name VARCHAR(200) NOT NULL,
    middle_name VARCHAR(200),
    last_name VARCHAR(200),

    display_name VARCHAR(500) NOT NULL,

    nationality VARCHAR(100),

    email CITEXT,
    phone VARCHAR(50),
    address TEXT,

    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

Privacy note

Kennitala er **ekki** í MVP schema.

Ef síðar verður krafa um hana myndi ég hafa hana í sérstöku vernduðu sviði/töflu, ekki bara:

```
kennitala VARCHAR(10)
í persons.
```

7. Organizations

```
CREATE TABLE organizations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

  name VARCHAR(500) NOT NULL,
  registration_number VARCHAR(50),

  organization_type VARCHAR(100),

  address TEXT,
  email CITEXT,
  phone VARCHAR(50),

  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

8. Parties

Par sem party getur verið annaðhvort `person` eða `organization`:

```
CREATE TABLE parties (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

  case_id UUID NOT NULL REFERENCES cases(id),

  person_id UUID REFERENCES persons(id),
  organization_id UUID REFERENCES organizations(id),

  role VARCHAR(50) NOT NULL,

  legal_representative_id UUID
    REFERENCES users(id),

  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

  CONSTRAINT parties_subject_chk
    CHECK (
```

```

                (person_id IS NOT NULL AND organization_id IS
NULL)
                OR
                (person_id IS NULL AND organization_id IS NOT
NULL)
            )

```

);

Þetta tryggir:

Party

|— Person

OR

|— Organization

aldrei bæði.

9. Events

```

CREATE TABLE events (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    event_date DATE NOT NULL,
    event_time TIME,

    title VARCHAR(500) NOT NULL,
    description TEXT,

    event_type VARCHAR(100),

    location VARCHAR(500),

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

Index:

```

CREATE INDEX idx_events_case_date
    ON events(case_id, event_date);

```

Þetta verður grundvöllur að **case timeline**.

10. Documents

```

CREATE TABLE documents (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    document_number VARCHAR(50),

    document_type_id UUID,

    title VARCHAR(500) NOT NULL,

    original_filename VARCHAR(1000),
    canonical_filename VARCHAR(1000),

    document_date DATE,
    received_date DATE,

    author VARCHAR(500),
    sender VARCHAR(500),
    recipient VARCHAR(500),

    source VARCHAR(500),

    mime_type VARCHAR(200),

    file_size BIGINT,

    ocr_status VARCHAR(50) DEFAULT 'NOT_REQUIRED',

    confidentiality_level_id UUID,

    legal_privilege BOOLEAN NOT NULL DEFAULT false,

    current_version_id UUID,

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    updated_by UUID REFERENCES users(id),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

11. Document versions

```

CREATE TABLE document_versions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    document_id UUID NOT NULL
        REFERENCES documents(id),

    version_number INTEGER NOT NULL,

    object_storage_key VARCHAR(2000) NOT NULL,

    sha256 CHAR(64) NOT NULL,

    file_size BIGINT NOT NULL,

    mime_type VARCHAR(200) NOT NULL,

    is_original BOOLEAN NOT NULL DEFAULT false,

    change_reason TEXT,

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    CONSTRAINT document_versions_unique
        UNIQUE(document_id, version_number)
);

```

Og síðan:

```

ALTER TABLE documents
ADD CONSTRAINT documents_current_version_fk
FOREIGN KEY (current_version_id)
REFERENCES document_versions(id);

```

12. Document relationships

```

CREATE TABLE document_relationships (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    source_document_id UUID NOT NULL
        REFERENCES documents(id),

    target_document_id UUID NOT NULL
        REFERENCES documents(id),

    relationship_type VARCHAR(50) NOT NULL,

```



```

        created_by UUID NOT NULL REFERENCES users(id),
        created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

        CONSTRAINT document_relationship_self_chk
            CHECK (source_document_id <> target_document_id)
    );

```

13. Evidence

```

CREATE TABLE evidence (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    document_id UUID REFERENCES documents(id),

    evidence_type VARCHAR(100),

    description TEXT,

    source VARCHAR(500),

    obtained_at TIMESTAMPTZ,
    obtained_by UUID REFERENCES users(id),

    custodian VARCHAR(500),

    original_format VARCHAR(100),

    integrity_status VARCHAR(50)
        NOT NULL DEFAULT 'UNKNOWN',

    relevance VARCHAR(50),

    admissibility_status VARCHAR(50),

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

14. Evidence custody

```

CREATE TABLE evidence_custody_events (

```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

evidence_id UUID NOT NULL
    REFERENCES evidence(id),

event_timestamp TIMESTAMPTZ NOT NULL
    DEFAULT now(),

action VARCHAR(100) NOT NULL,

from_user_id UUID REFERENCES users(id),
to_user_id UUID REFERENCES users(id),

location VARCHAR(500),

reason TEXT,

sha256 CHAR(64),

notes TEXT,

created_by UUID NOT NULL REFERENCES users(id),
created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

Pessi tafla á að vera **append-only**.

15. Issues

```

CREATE TABLE issues (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    title VARCHAR(500) NOT NULL,
    description TEXT,

    status VARCHAR(50) NOT NULL DEFAULT 'OPEN',

    priority VARCHAR(20) DEFAULT 'NORMAL',

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    resolved_at TIMESTAMPTZ

```

```
);
```

16. Facts

```
CREATE TABLE facts (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    case_id UUID NOT NULL REFERENCES cases(id),  
  
    statement TEXT NOT NULL,  
  
    status VARCHAR(50) NOT NULL DEFAULT 'UNKNOWN',  
  
    importance VARCHAR(20) DEFAULT 'NORMAL',  
  
    created_by UUID NOT NULL REFERENCES users(id),  
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

17. Legal arguments

```
CREATE TABLE legal_arguments (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    case_id UUID NOT NULL REFERENCES cases(id),  
  
    issue_id UUID REFERENCES issues(id),  
  
    title VARCHAR(500) NOT NULL,  
  
    argument_text TEXT NOT NULL,  
  
    position VARCHAR(50),  
  
    status VARCHAR(50) DEFAULT 'DRAFT',  
  
    created_by UUID NOT NULL REFERENCES users(id),  
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  
    updated_by UUID REFERENCES users(id),  
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

18. Many-to-many relationships

Fact ↔ Evidence

```
CREATE TABLE fact_evidence (  
    fact_id UUID NOT NULL REFERENCES facts(id),  
    evidence_id UUID NOT NULL REFERENCES evidence(id),  
  
    PRIMARY KEY (fact_id, evidence_id)  
);
```

Argument ↔ Fact

```
CREATE TABLE argument_facts (  
    argument_id UUID NOT NULL  
        REFERENCES legal_arguments(id),  
  
    fact_id UUID NOT NULL  
        REFERENCES facts(id),  
  
    PRIMARY KEY (argument_id, fact_id)  
);
```

Argument ↔ Legal Authority

```
CREATE TABLE argument_authorities (  
    argument_id UUID NOT NULL  
        REFERENCES legal_arguments(id),  
  
    authority_id UUID NOT NULL,  
  
    PRIMARY KEY (argument_id, authority_id)  
);
```

Star:

```
ALTER TABLE argument_authorities  
ADD CONSTRAINT argument_authorities_authority_fk  
FOREIGN KEY (authority_id)  
REFERENCES legal_authorities(id);
```

19. Legal authorities

```
CREATE TABLE legal_authorities (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

    authority_type VARCHAR(50) NOT NULL,

    title VARCHAR(1000) NOT NULL,

    citation VARCHAR(1000),

    authority_date DATE,

    court VARCHAR(500),

    reference VARCHAR(1000),

    url VARCHAR(2000),

    summary TEXT,

    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

20. Court proceedings

```

CREATE TABLE court_proceedings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    court VARCHAR(500) NOT NULL,

    court_case_number VARCHAR(100),

    court_level VARCHAR(100),

    jurisdiction VARCHAR(200),

    filing_date DATE,

    status VARCHAR(50),

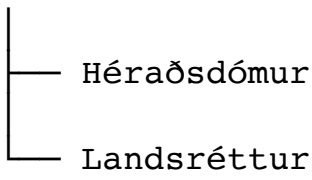
    judge VARCHAR(500),

    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

Pannig getum við haft:

CASE-001



Héraðsdómur

Landsréttur

sem tvö court_proceedings.

21. Court filings

```
CREATE TABLE court_filings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  proceeding_id UUID NOT NULL  
    REFERENCES court_proceedings(id),  
  
  document_id UUID NOT NULL  
    REFERENCES documents(id),  
  
  filing_type VARCHAR(100) NOT NULL,  
  
  filing_date TIMESTAMPTZ,  
  
  submitted_by UUID REFERENCES users(id),  
  
  submitted_to VARCHAR(500),  
  
  submission_method VARCHAR(100),  
  
  receipt_timestamp TIMESTAMPTZ,  
  
  court_status VARCHAR(100)  
    DEFAULT 'DRAFT',  
  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

22. Deadlines

```
CREATE TABLE deadlines (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  case_id UUID NOT NULL REFERENCES cases(id),  
  
  proceeding_id UUID
```

```

        REFERENCES court_proceedings(id),

title VARCHAR(500) NOT NULL,

description TEXT,

due_at TIMESTAMPTZ NOT NULL,

responsible_user_id UUID
        REFERENCES users(id),

status VARCHAR(50) NOT NULL DEFAULT 'OPEN',

source VARCHAR(500),

reminder_at TIMESTAMPTZ,

created_by UUID NOT NULL REFERENCES users(id),
created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

23. Legal holds

```

CREATE TABLE legal_holds (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    case_id UUID NOT NULL REFERENCES cases(id),

    reason TEXT NOT NULL,

    status VARCHAR(30) NOT NULL DEFAULT 'ACTIVE',

    created_by UUID NOT NULL REFERENCES users(id),
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    released_by UUID REFERENCES users(id),
    released_at TIMESTAMPTZ

);

```

Critical application rule:

```

ACTIVE legal hold
└──> DELETE = BLOCKED

```

24. Retention policies

```
CREATE TABLE retention_policies (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    name VARCHAR(500) NOT NULL,  
  
    retention_period_days INTEGER,  
  
    trigger_event VARCHAR(100),  
  
    approval_required BOOLEAN NOT NULL DEFAULT true,  
  
    legal_hold_required BOOLEAN NOT NULL DEFAULT true,  
  
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

25. Roles

```
CREATE TABLE roles (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    name VARCHAR(100) NOT NULL UNIQUE,  
  
    description TEXT  
);
```

Initial roles:

```
SYSTEM_ADMIN  
CASE_MANAGER  
LAWYER  
PARALEGAL  
REVIEWER  
EXTERNAL_COUNSEL  
READ_ONLY  
AUDITOR
```

26. Permissions

```
CREATE TABLE permissions (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```



```
resource VARCHAR(100) NOT NULL,  
  
action VARCHAR(100) NOT NULL,  
  
UNIQUE(resource, action)  
);  
Dæmi:
```

```
CASE / READ  
CASE / WRITE  
CASE / CLOSE
```

```
DOCUMENT / READ  
DOCUMENT / CREATE  
DOCUMENT / EXPORT
```

```
EVIDENCE / READ  
EVIDENCE / TRANSFER
```

```
AUDIT / READ
```

27. User roles

```
CREATE TABLE user_roles (  
    user_id UUID NOT NULL REFERENCES users(id),  
    role_id UUID NOT NULL REFERENCES roles(id),  
  
    PRIMARY KEY(user_id, role_id)  
);
```

28. Case permissions

```
CREATE TABLE case_permissions (  
    case_id UUID NOT NULL REFERENCES cases(id),  
  
    user_id UUID NOT NULL REFERENCES users(id),  
  
    permission VARCHAR(100) NOT NULL,  
  
    PRIMARY KEY(case_id, user_id, permission)  
);
```

Petta gefur okkur:

RBAC

+

Case-level ACL

sem er mun öruggara en RBAC eitt og sér.

29. Audit log

```
CREATE TABLE audit_log (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    event_timestamp TIMESTAMPTZ NOT NULL  
        DEFAULT now(),  
  
    user_id UUID REFERENCES users(id),  
  
    action VARCHAR(100) NOT NULL,  
  
    entity_type VARCHAR(100) NOT NULL,  
  
    entity_id UUID,  
  
    old_value JSONB,  
  
    new_value JSONB,  
  
    ip_address INET,  
  
    reason TEXT  
);
```

Ég myndi bæta við:

```
CREATE INDEX idx_audit_entity  
ON audit_log(entity_type, entity_id);
```

```
CREATE INDEX idx_audit_timestamp  
ON audit_log(event_timestamp);
```

```
CREATE INDEX idx_audit_user  
ON audit_log(user_id);
```

30. API design

API verður versioned:

/api/v1/

og JSON verður standard response format.

Dæmi:

GET /api/v1/cases/550e8400-e29b-41d4-a716-446655440000

31. Cases API

Create

POST /api/v1/cases

Request:

```
{
  "title": "Ágreiningur vegna verksamnings",
  "description": "Ágreiningur milli aðila...",
  "case_type": "CONTRACT_DISPUTE",
  "priority": "HIGH"
}
```

Response:

```
{
  "id": "uuid",
  "case_number": "LGM-2026-00123",
  "title": "Ágreiningur vegna verksamnings",
  "status": "NEW",
  "created_at": "2026-09-01T15:00:00Z"
}
```

Get

GET /api/v1/cases/{case_id}

Update

PATCH /api/v1/cases/{case_id}

Close

POST /api/v1/cases/{case_id}/close

Ég myndi **ekki** nota:

DELETE /api/v1/cases/{id}

fyrir venjulegan notanda.

32. Case overview endpoint

Mjög gagnlegur endpoint:

GET /api/v1/cases/{case_id}/overview

Return:

```
{
  "case": {},
  "parties": [],
  "events": [],
  "issues": [],
  "documents": [],
  "evidence": [],
  "deadlines": [],
  "proceedings": [],
  "active_legal_holds": []
}
```

Þetta verður líklega aðal endpoint fyrir case dashboard.

33. Documents API

Upload

POST /api/v1/cases/{case_id}/documents

Content-Type: multipart/form-data

Pipeline:

UPLOAD

↓

MALWARE SCAN

↓

HASH

↓

METADATA

↓

OCR

↓

STORE

↓

INDEX

↓

AUDIT

Get metadata

GET /api/v1/documents/{document_id}

Download

GET /api/v1/documents/{document_id}/download

Þessi endpoint ætti að skila **temporary signed URL** eða streama binary gegnum authorization layer.

Ekki:

/public/documents/abc.pdf

34. Document versions API

GET /api/v1/documents/{id}/versions

GET /api/v1/documents/{id}/versions/{version_id}

POST /api/v1/documents/{id}/versions

Original version:

READ ONLY

35. Evidence API

POST /api/v1/cases/{case_id}/evidence

GET /api/v1/evidence/{evidence_id}

POST /api/v1/evidence/{evidence_id}/custody-events

Dæmi:

```
{
  "action": "TRANSFERRED",
  "to_user_id": "uuid",
  "reason": "Sent to external counsel"
}
```

36. Timeline API

GET /api/v1/cases/{case_id}/events

POST /api/v1/cases/{case_id}/events

Mögulegt query:

GET /api/v1/cases/{id}/events?from=2026-01-01&to=2026-09-01

37. Issues / Facts / Arguments

GET /api/v1/cases/{id}/issues

POST /api/v1/cases/{id}/issues

GET /api/v1/issues/{id}

PATCH /api/v1/issues/{id}

Facts:

GET /api/v1/cases/{id}/facts

POST /api/v1/cases/{id}/facts

Arguments:

GET /api/v1/cases/{id}/arguments

POST /api/v1/cases/{id}/arguments

38. Court API

POST /api/v1/cases/{id}/proceedings

GET /api/v1/cases/{id}/proceedings

GET /api/v1/proceedings/{id}

POST /api/v1/proceedings/{id}/filings

39. Court submission validation

Ég myndi hafa sérstakt endpoint:

POST /api/v1/proceedings/{id}/validate-submission

Response:

```
{
  "valid": false,
  "errors": [
    {
      "document": "DOC-123",
      "code": "NOT_SEARCHABLE",
      "message": "PDF is not searchable"
    }
  ],
  "warnings": [
    {
      "document": "DOC-456",
      "code": "FILENAME"
    }
  ]
}
```

Þetta er mjög áhugaverður eiginleiki fyrir íslenskt kerfi vegna þeirra forma- og afhendingarkrafna sem Dómstólasýslan setur fyrir skjöl sem lögð eru fram.

40. Search API

GET /api/v1/search

Dæmi:

GET /api/v1/search?q=vanefnd+verksamningur

Advanced:

```
GET /api/v1/search
    ?q=vanefnd
    &case_id=uuid
    &document_type=EMAIL
    &from=2026-01-01
    &to=2026-08-31
```

Response:

```
{
  "total": 42,
  "results": [
    {
      "type": "document",
      "id": "uuid",
      "title": "Tölvupóstur vegna samnings",
      "case_id": "uuid",
      "score": 12.43
    }
  ]
}
```

41. API security

Allir endpoints nema health checks:

Authentication

↓

Authorization

↓

Case permission

↓

Resource permission

↓

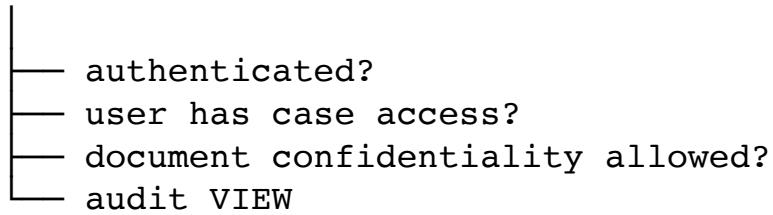
Action

↓

Audit

Dæmi:

GET document



42. HTTP status codes

Við notum hefðbundna merkingu:

200 OK

201 Created

204 No Content

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

422 Validation Error

423 Locked

429 Too Many Requests

500 Internal Server Error

423 Locked gæti verið gagnlegt þegar resource er læst vegna legal hold/workflow.

43. Error format

Allar villur með sama structure:

```
{
  "error": {
    "code": "DOCUMENT_NOT_SEARCHABLE",
    "message": "Document cannot be submitted because it is
not searchable.",
    "details": {}
  }
}
```

Ekki senda stack traces til notanda.

44. API pagination

Allar stórar collections:

GET /api/v1/cases/{id}/documents?limit=50&offset=0
eða cursor-based pagination síðar:

GET /api/v1/cases/{id}/documents?limit=50&cursor=...
Ég myndi velja **cursor pagination** þegar kerfið stækkar.

45. API filtering / sorting

Dæmi:

```
GET /api/v1/cases
    ?status=OPEN
    &priority=HIGH
    &sort=-opened_at
    &limit=50
```

46. API idempotency

Fyrir aðgerðir sem geta verið endurteknaðar vegna network failures:

Idempotency-Key: 8f2c...
Sérstaklega:

```
document upload
court filing
evidence transfer
external submission
```

Þetta kemur í veg fyrir tvítekningu.

47. Transactions

Dæmi:

Þegar document er búið til:

```
BEGIN
```

```
create document
create version
create audit event
```

```
update current_version
```

COMMIT

Ef eitthvað bilar:

ROLLBACK

Object storage er þó ekki transactional á sama hátt og PostgreSQL. Þess vegna þarf **outbox/background-job pattern** fyrir flóknari workflow.

48. Outbox

Ég myndi bæta þessu við í v0.3:

```
CREATE TABLE outbox_events (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  event_type VARCHAR(200) NOT NULL,  
  
  aggregate_type VARCHAR(100) NOT NULL,  
  aggregate_id UUID NOT NULL,  
  
  payload JSONB NOT NULL,  
  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  
  processed_at TIMESTAMPTZ  
);
```

Þetta getur síðan keyrt:

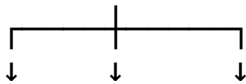
Document created



Outbox event



Worker



OCR Index Notification

49. API documentation

FastAPI getur sjálfkrafa framleitt:

OpenAPI 3.x

og:

/api/docs

/api/redoc

Þetta verður síðan grunnur fyrir:

- frontend
- integration tests
- external integrations
- API client generation.

50. API resource hierarchy

Heildar API verður í grófum dráttum:

/api/v1

/cases

- /{case_id}
 - /parties
 - /documents
 - /events
 - /issues
 - /facts
 - /arguments
 - /evidence
 - /deadlines
 - /proceedings
 - /legal-holds
 - /overview

/documents

- /{document_id}
 - /versions
 - /download
 - /relationships

/evidence

- /{evidence_id}
 - /custody-events

/issues

/facts

/arguments

/legal-authorities

```
/proceedings
  /{proceeding_id}
    /filings
    /validate-submission
```

```
/users
/search
/audit
```

51. Mikilvæg breyting frá v0.2

Við höfum nú í raun fundið eitt atriði sem ég myndi bæta við upphaflega hönnunina:

Outbox Events

Þetta verður mikilvægt þegar við förum að vinna með:

- OCR
- AI
- search indexing
- email ingestion
- notifications
- document processing
- integrations.

Þannig helst **PostgreSQL transaction** aðalckerfið og bakgrunnsþjónustur verða ekki beint að fíkta í kjarnagögnum.

52. Heildar gagnagrunnurinn

Við erum nú með eftirfarandi:

CORE

```
users
persons
organizations
cases
parties
```

DOCUMENT

documents
document_versions
document_relationships

CASE ANALYSIS

events
issues
facts
legal_arguments
legal_authorities

EVIDENCE

evidence
evidence_custody_events
fact_evidence
argument_facts
argument_authorities

COURT

court_proceedings
court_filings
deadlines

SECURITY

roles
permissions
user_roles
case_permissions
audit_log

RETENTION

legal_holds
retention_policies

INTEGRATION

outbox_events
27 töflur í þessu stigi.

53. Næsta stig: v0.4

Ég myndi nú **ekki** fara beint í að skrifa alla SQL migration-kóðana. Það vantar eitt mikilvægt millistig.

Technical Specification v0.4 — Workflow & Security Model

Þar myndum við skilgreina nákvæmlega:

Case lifecycle
Document lifecycle
Evidence lifecycle
Court filing lifecycle
Approval workflow
Legal hold workflow
Retention workflow
User / role model
Case access model
Audit model
Til dæmis:

DOCUMENT

Uploaded
↓
Scanning
↓
Validated
↓
OCR
↓
Classified
↓
Human Review
↓
Approved
↓
Active
↓
Archived
og:

COURT FILING

Draft

↓
Legal Review

↓
Approved

↓
Submitted

↓
Received

↓
Filed

Þetta skref er mikilvægt áður en við skrifum endanlegt PostgreSQL schema, því workflow-kröfurnar munu ákvarða hvaða state-töflur, history-töflur, permissions og database triggers við þurfum.

Eftir v0.4 getum við farið í **v0.5 – Production PostgreSQL Schema**, þar sem ég myndi skila fullri `001_initial_schema.sql`, með foreign keys, indexes, triggers, audit enforcement, seed data og migration strategy.